



Shared functional connectome fingerprints following ritualistic ayahuasca intake

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ABSTRACT

The knowledge that brain functional connectomes are unique and reliable has enabled behaviourally relevant inferences at a subject level. However, whether such “fingerprints” persist under altered states of consciousness is unknown. Ayahuasca is a potent serotonergic psychedelic which produces a widespread dysregulation of functional connectivity. Used communally in religious ceremonies, its shared use may highlight relevant novel interactions between mental state and functional connectome (FC) idiosyncrasy. Using 7T fMRI, we assessed resting-state static and dynamic FCs for 21 Santo Daime members after collective ayahuasca intake in an acute, within-subject study. Here, connectome fingerprinting revealed FCs showed reduced idiosyncrasy, accompanied by a spatiotemporal reallocation of keypoint edges. Importantly, we show that interindividual differences in higher-order FC motifs are relevant to experiential phenotypes, given that they can predict perceptual drug effects. Collectively, our findings offer an example of how individualised connectivity markers can be used to trace a subject’s FC across altered states of consciousness.

1. Introduction

The uniqueness of one’s brain connectivity profile is being increasingly recognised as a ubiquitous principle of *connectomics*. Akin to the ridges and furrows that comprise our fingerprints, functional connectivity patterns derived from functional resonance magnetic imaging (fMRI) data, also known as functional connectomes (FCs) (Smith et al., 2013), have been found to be stable across a lifetime (Horien et al., 2019) and hold explanatory power for robust inferences at a single-subject level (Shen et al., 2017). Evidence has shown complex behavioural phenotypes such as cognition (Sripada et al., 2020), demographics (Nielsen et al., 2019), traits such as fluid intelligence (Li et al., 2020) or personality (Dubois et al., 2018), and even clinical outcomes (Abdallah et al., 2021) can be reliably predicted from FCs alone. This observation has led to calls to move away from group-level inferences and towards interindividual differences prior to concluding

on the generalisability of brain activity (Seghier and Price, 2018; Dubois and Adolphs, 2016).

In recent years, efforts have been underway to develop the field of “brain fingerprinting” (Amico and Goñi, 2018). First exemplified by Finn et al., individual subjects were shown to be readily distinguishable from a set of FCs based on their correspondence (Finn et al., 2015). Since then, work has demonstrated that an individual’s connectome fingerprint across sessions can be separated into signalling motifs reflecting both trait intra-subject and state-dependent inter-subject variance (E. Sareen et al., 2021; Geerligs et al., 2015), reproducible across modalities (Elliott et al., 2019; Vanderwal et al., 2017), acquisition methods (de Souza Rodrigues et al., 2019; Hakim et al., 2021; Wu et al., 2022) and durations (Amico and Goñi, 2018; Airan et al., 2016; da Silva Castanheira et al., 2021). These findings have contributed to the notion that, across mental states, there exists an “intrinsic” functional network architecture which is inherent to brain function and exhibits subtle

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variations among individuals (Fox and Raichle, 2007; Vincent et al., 2007; Gratton et al., 2018). That said, it is important to consider that these fingerprints of brain organisation might not just be limited to the spatial organisation and independence of FC traits but likely also to their temporal quality (Van De Ville et al., 2021). Spatiotemporal dynamics have been suggested to provide a "common currency" for mental and neuronal states (Northoff et al., 2020), with neural processing being organised across timescales and increasing along the cortical hierarchy of information processing (Raut et al., 2020). According to this view, shorter timescales in sensory areas facilitate the rapid detection and encoding of dynamic stimuli, which are subsequently integrated by the slower dynamics of associative areas over longer timeframes (Hasson et al., 2008; Golesorkhi et al., 2021).

Much work has been concerned with understanding how such inherent connectivity might be differentially altered according to a particular individual or mental state (Elliott et al., 2019; Gratton et al., 2018; Porter et al., 2022). However, there is little evidence bridging these lines of research, particularly quantifying the variance associated with a subject versus the brain state under which it is examined (Finn et al., 2017). Compelling evidence reveals agonism of the 5-HT_{2A} receptor by serotonergic psychedelics holds a central role in shaping altered states of consciousness (ASCs) (Vollenweider and Preller, 2020) and thus, potentially connectome fingerprints. Whole-brain modelling has implicated 5-HT_{2A} receptor distribution in shaping brain dynamics (Singleton et al., 2022) whereas its stimulation enhances the temporal diversity of brain activity (Herzog et al., 2023). These effects yield downstream shifts in functional coupling between large-scale networks, ultimately diminishing integrative processing across major brain networks (M. K. Doss et al., 2021). Within a hierarchical predictive processing framework, these outcomes are hypothesised to be linked to the induced subjective experience via the decreased confidence in priors encoded by functional hierarchies (Carhart-Harris and Friston, 2019). Accordingly, it may therefore be the case that stimulation of 5-HT_{2A} receptors could perturb behaviourally relevant brain fingerprints otherwise residing within high-order functional networks (Mantwill et al., 2022). Indeed, classical psychedelics have been speculated to be potential therapeutic interventions by improving symptomatology through rebalancing aberrant brain states (Carhart-Harris and Friston, 2019; M. K. Doss et al., 2021; Daws et al., 2022). To date however, it is unknown how different classical psychedelics might alter connectome fingerprints. Subject-level analyses as devised by fingerprinting may therefore prove to be best-suited for modelling the neurobiology of 5-HT_{2A} agonists, given their highly heterogeneous subjective experiences, plasma drug concentrations, and divergent effects on resting-state network organisation (Moujaes et al., 2023).

A relevant practice that is purported to achieve a communal ASC is the ritualistic use of the psychedelic brew ayahuasca. Devised from a combination of two different plant sources, the vine *Banisteriopsis caapi* and *Psychotria viridis*, ayahuasca produces a profound change to subjective experience, comprising a diffuse state of cognition alongside complex changes to self-referential awareness, perception, and mood (Riba et al., 2001). Whereas *Psychotria viridis* is a rich source of the potent 5-HT_{2A} agonist N,N-dimethyltryptamine (DMT), *Banisteriopsis caapi* contains monoamine oxidase inhibitor (MAOI) β -carbolines such as harmine, harmaline, and tetrahydroharmine, serving to promote the bioavailability of DMT (Riba et al., 2003). Historically, ayahuasca is used by syncretic religions such as *Santo Daime* to achieve personal insight, intimacy and spiritual development (Lowell and Adams, 2017). Members of the congregation drink ayahuasca (termed *Daime*) communally in a ceremony referred to as the "works" (*trabalhos*). These are collective endeavours performed by members of the congregation consisting of alternating periods of song, dance, and attentive silence. Providing a formalised type of set and setting, members follow a prescribed mental state with which to engage their symbolic and religious framework (*doctrina*) (Hartogsohn, 2021). This ritualistic use of ayahuasca might therefore provide a useful means by which to

investigate the dissimilarity between trait and state FC under conditions in which an individual transitions from a normal, waking state of consciousness to a shared altered state.

Here, we sought to understand how the inherency of a subject's FC might alter under the altered state of consciousness induced by the ritualistic consumption of ayahuasca brew. Replicating a brain fingerprint framework (Van De Ville et al., 2021) in *Santo Daime* members, we characterised changes to both static and dynamic connectome identifiability at peak drug effects. Furthermore, we explored how changes to an individual's underlying functional connectivity might subsequently help explain aspects of their subjective experience.

2. Methods

2.1. Participants

Twenty-four volunteers were enrolled in a within-subject, fixed-order observational study. Data from three volunteers were excluded from analyses due to excessive head motion leaving a final sample of 21 subjects (10 females) of ages 29 to 64 (M : 54.48, SD : 10.55). The cohort consisted of experienced members of the Dutch chapter of the church of *Santo Daime*. Individuals were selected based on an exclusion criterion comprising the absence of ferromagnetic devices/implants (MRI contraindications), pregnancy and use of (medicinal) substances in the past 24 h. Detailed demographic information pertaining to the final sample can be found in Table S1.

All participants were fully informed of all procedures, possible adverse reactions, legal rights and responsibilities, expected benefits, and their right to voluntary termination without consequences. The study was conducted according to the Declaration of Helsinki (1964) and amended in Fortaleza (Brazil, October 2013) and in accordance with the Medical Research Involving Human Subjects Act (WMO) and was approved by the Academic Hospital and University's Medical Ethics committee of Maastricht University (NL70901.068.19/METC19.050).

2.2. Study procedures

Participants underwent two consecutive test days; one baseline condition (sober) followed by an acute condition under the influence of ayahuasca as reported previously (Ramaekers et al., 2023) and outlined in Fig. 1. Participants self-administered a volume of ayahuasca equivalent to their usual dose (mean 24 ml, SD : 8.16), prepared from a single batch by the Church of *Santo Daime* and analysed according to prior referencing standards (see Supplementary). The brew used contained 0.14 mg/ml of DMT, 4.50 mg/ml of harmine, 0.51 mg/ml of harmaline, and 2.10 mg/ml of tetrahydroharmine. Each self-administration took place during a ceremony organised and supervised by the *Santo Daime* church. As to facilitate the communal use of ayahuasca, participants drank the ayahuasca brew individually in the company of fellow members while performing their collective works (singing, dancing, meditation). Participant dosing schedules were stratified across each lab visit with testing performed within 4 pairs of visits (6 subjects per cycle) with each subject being tested at the same window of time as to minimise diurnal variation. The research team was not involved in the organisation of the ceremonies nor the production, dosing, or administration of ayahuasca.

On each day upon arrival to the lab, the absence of drug and alcohol use was assessed via a urine drug screen and a breath alcohol test. An additional pregnancy test was given if the participants were female. Each visit consisted of a 30-minute wait period, followed by a 1 h MRI scanning session occurring 1 h after intake. On day 2, venous blood samples were collected approximately 60 and 160 min after ayahuasca intake to assess serum concentrations of alkaloids according to laboratory protocols (see Supplementary). The retrospective 5-Dimensions of Altered States of Consciousness (5D-ASC) scale (Studerus et al., 2010) and the Ego Dissolution Inventory (Nour et al., 2016) were administered

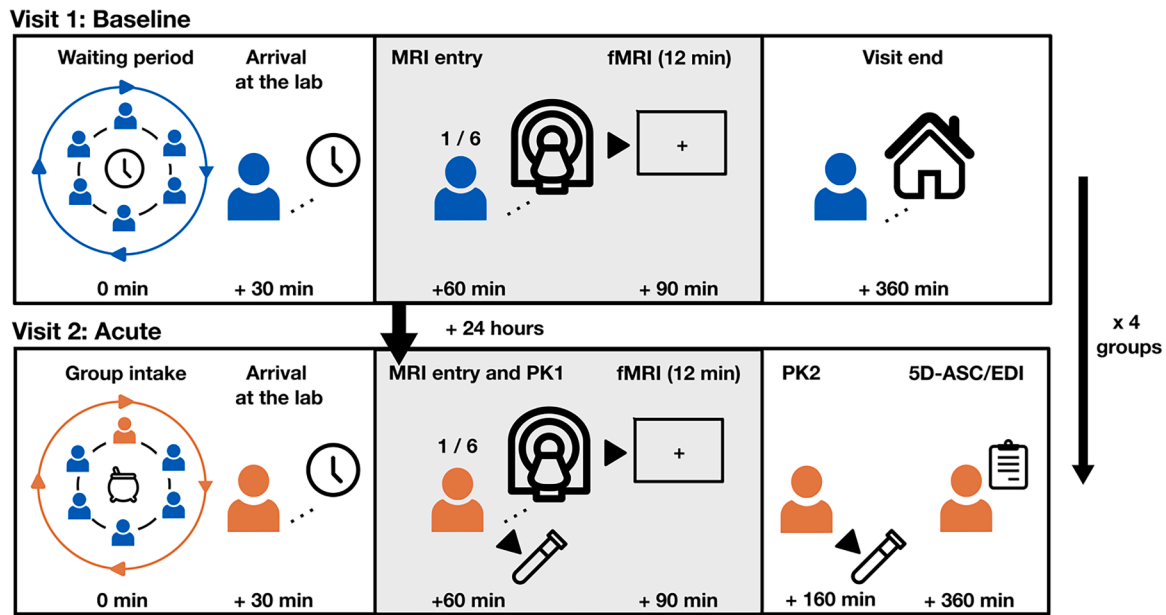


Fig. 1. Testing day schedule. Overview of each visit to the lab. 4 groups of 6 Santo Daime members visited the lab on two occasions. Acute dosing days comprised of successive self-administrations in the presence of other group members. Participants were then sequentially scanned after their respective intake, with testing timelines following the same schedule as the preceding baseline visit to minimise diurnal variation. Acute dosing visits comprised additional serum pharmacokinetic measurements at +60 min and +160 min. Retrospective questionnaires (Ego Dissolution Inventory [EDI], 5-Dimensions of Altered States Questionnaire [5D-ASC]) were administered at the end of each acute visit.

360 min after drug ingestion to assess the subjective experience after drug intake. Following study completion, each subject was contacted for an online follow-up (+ 6 months). In order to gauge the prevalence of mental processes pertaining to the ceremonial use of Daime during resting-state, participants were asked to answer visual analogue scales (0–100) inquiring as to whether they were internally *singing* or employing *meditation* in the scanner. Given the time delay, questions pertaining to their *recollection* of each resting state acquisition were also provided. For more information regarding all inventories, see the Supplementary.

2.3. Image acquisition

Images were acquired on a MAGNETOM 7T MRI scanner. On each visit, participants underwent a structural MRI (60 min post-treatment), single-voxel proton MRS in the PCC (70 min post) and visual cortex (80 min post), and fMRI (90 min post), during peak subjective effects. Findings and methods pertaining to MRS are to be reported elsewhere.

T1-weighted anatomical images were acquired using a magnetisation-prepared 2 rapid acquisition gradient-echo (MP2RAGE) sequence (TR = 4500 ms, TE = 2.39 ms, TI1 = 0.90 s, TI2 = 2.75 s, flip angle 1 = 5°, flip angle 2 = 3°, voxel size = 0.9 mm isotropic, matrix size = 256 × 256 × 192, phase partial Fourier = 6/8, GRAPPA = 3 with 24 reference lines, bandwidth = 250 Hz/pixel). 500 echo planar imaging (EPI) volumes were acquired at rest (TR = 1400 ms; TE = 21 ms; field of view = 198 mm; flip angle = 60°; oblique acquisition orientation; interleaved slice acquisition; 72 slices; slice thickness = 1.5 mm; voxel size = 1.5 × 1.5 × 1.5 mm) followed by 5 phase encoding volumes acquired for EPI unwarping. Shimming was performed with fixed criterion for all scans, with dorsal regions prioritised and the cerebellum excluded. Partial coverage of the temporal pole (< 100 voxels) arose for 4/42 scans. During EPI acquisition, participants were shown a black fixation cross on a white background.

2.4. Functional pre-processing

All pre-processing steps were performed according to an in-house

pipeline (Amico et al., 2020; Amico et al., 2017) based on FSL (FMRIB software library, FSL 6.0; www.fmrib.ox.ac.uk/fsl) and implemented in MATLAB (R2019b). The individual functional connectomes (FCs) were modelled in the native BOLD fMRI space of each subject.

MP2RAGE images were first denoised to improve signal-to-noise ratio (Choi et al., 2019), bias-field corrected (FSL FAST), skull-stripped (HD-BET) (Isensee et al., 2019), and then segmented (FSL FAST) to extract white matter, grey matter and cerebrospinal fluid (CSF) tissue masks. These masks were warped in each individual subject's functional space by means of subsequent linear and non-linear registrations (FSL *flirt* 6dof, FSL *flirt* 12dof and FSL *fnirt*). BOLD fMRI volumes were pre-processed in line with Power et al. (Power et al., 2014; Power et al., 2012). Subsequent steps included: deletion of 2 initial volumes (FSL *utils*), slice timing correction (FSL *slcrtimer*), BOLD volume unwarping (FSL *topup*), realignment (FSL *mcflirt*), normalization to mode 1000, demeaning and linear detrending (Matlab *detrend*), regression (Matlab *regress*) of 18 signals: 3 translations, 3 rotations, and 3 tissue-based regressors (mean signal of wholebrain [global signal], white matter [WM] and cerebrospinal fluid [CSF]), as well as 9 corresponding derivatives (backwards difference; Matlab). A bandpass first-order Butterworth filter [0.009 Hz, 0.08 Hz] was then applied to all BOLD timeseries at the voxel level (Matlab *butter* and *filtfilt*). As an additional cleaning step, the first three principal components of the BOLD signal in the WM and CSF tissue were subsequently regressed out of the grey matter (GM) signal (Matlab, *pca* and *regress*) at the voxel level, in line with the aCompCor methodology described by Behzadi et al. (Behzadi et al., 2007). These principal components were included as nuisance parameters within a general linear model for the BOLD time series data extraction. No smoothing was performed.

We also kept track of the fMRI volumes that were highly influenced by head motion, by using three different metrics as a scrubbing index: 1) Frame Displacement (FD, in mm); 2) DVARS (D referring to temporal derivative of BOLD time courses, VARS referring to root mean square variance over voxels) (Power et al., 2014); 3) SD (standard deviation of the BOLD signal within brain voxels at every time-point. Outlier BOLD volumes were defined as having a 1) FD > 0.5; 2) DVARS > 75 percentile + 1.5 of the interquartile range; 3) SD > 75 percentile + 1.5 of

the interquartile range. It should be noted no volume censoring was performed using this index. Rather, this information was used as a confound in our multilinear regression analyses and quality control assessments (see Fig. 6 and Figure. S2). Functional connectomes obtained with and without scrubbing were highly similar (average Pearson's $r = 0.99$) with no significant differences in motion being identified between or within conditions (see Figure. S2).

A 2 mm cortical Schaefer parcellation (Schaefer et al., 2017) based on 200 brain regions (publicly available at: https://github.com/ThomasYeoLab/CBIG/tree/master/stable_projects/brain_parcellation/Schaefer2018), was projected into each subject's T1 space (FSL flirt 6dof and FSL flirt 12dof) and subsequently their native EPI space. FSL boundary-based-registration was also applied to improve the registration of the structural masks and parcellation to the functional volumes. Regions of interest (ROIs) were ordered according to seven cortical RSNs as proposed by Yeo et al. (Yeo et al., 2011). These included the visual (VIS), somatomotor (SM), dorsal attention (DA), ventral attention (VA), limbic (L), frontoparietal (FP) and the default mode network (DMN).

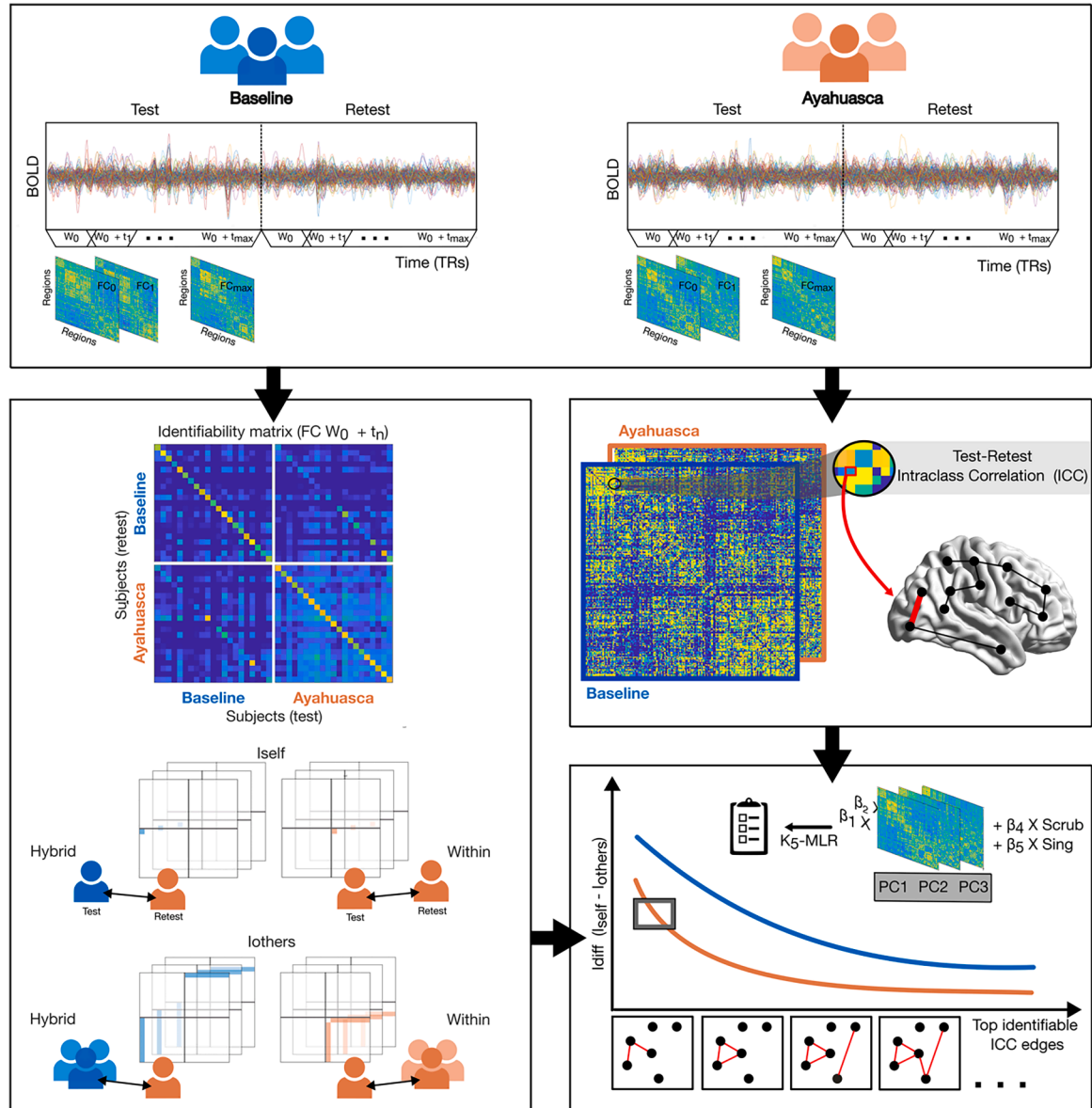


Fig. 2. Connectome fingerprinting workflow. First, each subject fMRI timeseries is split into test vs retest halves. For all subjects and conditions (baseline, ayahuasca), dFC frames are computed for increasing temporal windows until t_{max} is reached. Connectome fingerprints can next be calculated as the similarity in functional connectivity for all combinations of FC test vs retest (within condition and between), yielding an identifiability matrix (Wu et al., 2022) per timescale (left). Each colour matched block reflects identifiability *within* a condition whereas colour mismatched (“hybrid”) blocks represent the distance of each subject’s identifiability *between* conditions. This object allows us to compute for each subject: I_{self} (represented by each diagonal element) denoting their similarity to oneself and I_{others} (represented by the average of each row/column) representing their similarity to others, for both *within* and *between* conditions. In parallel, we can assess the fingerprinting value of specific edges per condition and timescale by calculating their intraclass correlation coefficient (ICC, right). We can next rank edges according to their ICC and iteratively calculate a compound measure of I_{self} and I_{others} (I_{diff}). This allows us to examine how edges “driving” one’s fingerprint evolve under Ayahuasca. Lastly, we can assess their experiential relevance by fitting an iterative multi-linear model comprising PCA-derived principal components (PCs) of their functional connectivity as predictors of interest. Decomposing their signal into components ranked according to explained variance (PC1–3), the relevance of cohort-level (high-variance – PC1) and individual-level functional connectivity information (low-variance – PC3) is simultaneously assessed, while accounting for motion and “working”. At each step, model performance and generalisability are measured using k-fold (K_5) cross validation, yielding an optimal edge cut-off.

2.5. Assessment of functional connectivity

To assess connectome fingerprints (described in our next section) across static and dynamic temporal scales, we devised two separate workflows for functional connectivity. Static functional connectivity between each pair of ROIs (edges) was calculated with a Pearson correlation coefficient between each pair of mean signal time courses (across a single run). For each subject, this results in an $N \times N$ FC matrix, where N is the number of ROIs, and with each element in the FC representing the connectivity strength between a pair of ROIs. Secondly, we assessed the dynamic functional connectomes (dFC) by performing a sliding window analysis to produce sets of connectivity matrices reflecting the temporal development of whole-brain functional connectivity (across our 249 timepoints). We captured relevant FC patterns by balancing the number of time points for a stable dFC computation, exploring sets of dFCs across 5 different window lengths of: 70 s, 140 s, 210 s, 280 s and 349 s. Each window step was fixed to 14 s, the equivalent of 10 TRs as per prior work (Van De Ville et al., 2021).

2.6. Whole-brain connectome identifiability

Changes to the identifiability of each subject's functional connectome were quantified by replicating the methodology originally proposed by Amico et al., devised for both static and dynamic functional connectivity (Amico and Goñi, 2018; Van De Ville et al., 2021).

The approach devises an identifiability matrix for each condition, consisting of a matrix of correlations (Pearson, square, non-symmetric) between a subject's test and retest functional scans. We firstly split each scan into two corresponding halves (249 vol each, or 6 min) to generate test-retest sets for each condition. Prior work has shown fMRI scan lengths of 3 min are sufficient to produce reliable fingerprints (Amico and Goñi, 2018; Airan et al., 2016). Since connectivity matrices are

symmetric, we can then extract unique elements of each test-retest FC by taking the upper triangle of each matrix; resulting in a $1 \times 19,900$ vector of edge values for each subject per condition which can then be compared using Pearson correlation, either between different subjects in the same condition or within the same subject across conditions. This yields the “identifiability matrix” as outlined in Fig. 2.

2.6.1. Static identifiability

In the case of static FCs, let $\mathbf{A}$ be the “identifiability matrix”, between the subjects' FCs test and retest. The dimension of $\mathbf{A}$ is N^2 , where N is the number of subjects in the study. In its original formulation (Amico and Goñi, 2018), let $I_{\text{self}} = \langle a_{ii} \rangle$ represent the average of the main diagonal elements of $\mathbf{A}$, which consist of the Pearson correlation values between test-retest sets of same subjects, otherwise defined as self-identifiability or I_{self} . Similarly, let $I_{\text{others}} = \langle a_{ij} \rangle$ define the average of the off-diagonal elements of matrix $\mathbf{A}$, i.e., the correlation between test-retest sets of different subjects. Lastly, let the differential identifiability (I_{diff}) of the population be the difference between both terms, otherwise denoted as:

$$I_{\text{diff}} = (I_{\text{self}} - I_{\text{others}}) \cdot 100$$

Which provides an indication of the difference between the average within-subject FCs similarity and the average between-subjects FCs similarity. The greater the I_{diff} the higher the individual fingerprinting value (or “uniqueness”) across the sample. In the present study, by extrapolating each individual element of $\mathbf{A}$ (see Fig. 2) we can define I_{diff} , I_{self} and I_{others} scores per subject. As an additional step, we also sought to derive the distance of each participant fingerprint under ayahuasca (i.e., I_{self} I_{others} I_{diff}) from their respective normative state from their baseline. Using the approach outlined by Sorrentino et al. for static connectomes (Sorrentino et al., 2021) we calculated the identifiability matrices across combinations of different conditions (e.g., the Pearson correlation of test-sober, retest-ayahuasca). When concatenated

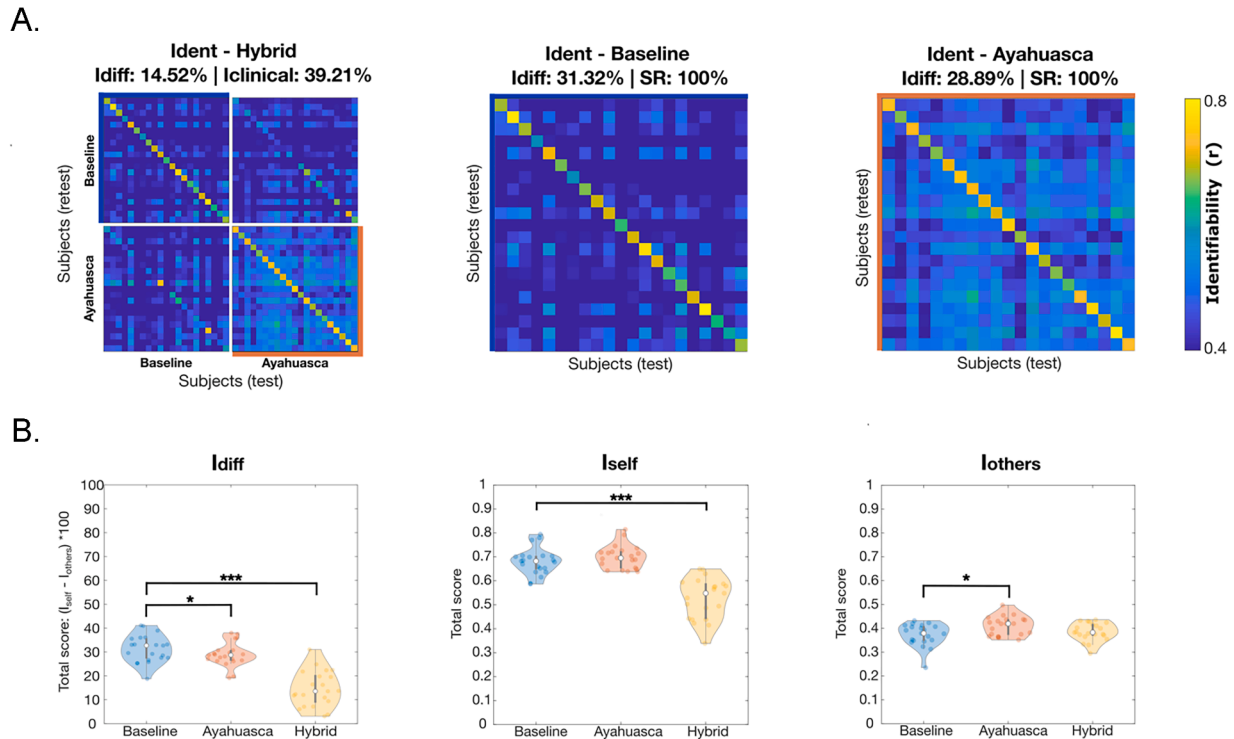


Fig. 3. Whole-brain measures of static identifiability. (A) shows the identifiability matrix (far left) at T_{max} with corresponding “standard” identification matrices for each condition expanded on the right. From hybrid off-block elements one can also define the I_{clinical} for a participant as the average similarity of the individual connectome of a subject with respect to the baseline. For all, differential identifiability (I_{diff}) values and success rates (SRs, where applicable) on top also provide complementary scores of the fingerprint level (see Methods). (B) Violin plots highlighting the difference of each identifiability metric (I_{diff} , I_{self} , I_{others}) between conditions. Hybrid counterparts are also presented in respect to baseline. Two-tail significance is denoted as follows: $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

with our within-group identifiability matrices this produces a hybrid identifiability matrix (see Figs. 2 and 3), where the between blocks (groups) elements and scores reflect the similarity (or distance) between the test-retest connectomes of subjects across different conditions. By averaging, this also allows us to derive a final overall cohort I_{clinical} score which provides a percentage (average) score of how similar their connectome with respect to baseline is (0% - totally dissimilar, 100% totally equivalent) across test-retest splits. Finally, we also measured the success-rate (SR) of the identification procedure as percentage of cases with higher I_{self} vs I_{others} (Finn et al., 2015; E. Sareen et al., 2021). For completeness, we calculated per condition the significance of both observed I_{diff} and SR scores in respect to their null equivalents using permutation testing (see Supplementary).

2.6.2. Dynamic identifiability

We can next extend this principle to dynamic functional connectomes (dFC) by calculating each measure across each dynamic frame of connectivity (see Fig. 2 for an overview). For a fixed window length w , the resulting dynamic identifiability matrix is then a block diagonal matrix, where each block represents the self-similarity within the dFC frames of a specific subject. The off-diagonal blocks, in this representation, encode instead the between-dFC frames similarity across different subjects (dynamic I_{others}). Let $S_{M_T} = \{\text{dFC}_1, \text{dFC}_2, \dots, \text{dFC}_N\}$ be the set of dFC frames in the test session for a specific subject M . Similarly, let $S_{M_{RT}}$ represent the set of dFC frames in the retest session for the same subject M . We can then define the dynamic I_{self} (dI_{self}) for subject M as:

$$dI_{\text{self}}(M) = \frac{\sum_{i \in S_{M_T}} \sum_{j \in S_{M_{RT}}} \text{corr}(dFC_i, dFC_j)}{|S_{M_T}| \cdot |S_{M_{RT}}|} = \frac{\sum_{i \in S_{M_T}} \sum_{j \in S_{M_{RT}}} dI_{ij}}{|S_{M_T}| \cdot |S_{M_{RT}}|}$$

Where by $|S_{M_T}|$, $|S_{M_{RT}}|$ define the cardinalities of the sets. Similarly, let $|S_{F_T}|$, $|S_{F_{RT}}|$ define the sets for a different subject F . We can define dynamic I_{others} (dI_{others}) as:

$$dI_{\text{others}}(M, F) = \frac{1}{2} \left(\frac{\sum_{i \in S_{M_T}} \sum_{j \in S_{F_{RT}}} dI_{ij}}{|S_{M_T}| \cdot |S_{F_{RT}}|} + \frac{\sum_{i \in S_{F_T}} \sum_{j \in S_{M_{RT}}} dI_{ij}}{|S_{F_T}| \cdot |S_{M_{RT}}|} \right)$$

and hence:

$$dI_{\text{others}}(M) = \frac{\sum_{F=1}^S dI_{\text{others}}(M, F)}{S - 1}, F \neq M$$

where the summation is over the total number of subjects S other than M . Last, dI_{diff} for a subject M can be described as:

$$dI_{\text{diff}}(M) = (dI_{\text{self}}(M) - dI_{\text{others}}(M)) * 100$$

2.7. Edgewise connectome identifiability

In order to understand which edges key contributors were to changes in connectome identifiability, we quantified the edgewise reliability for each brain region pair and every test-retest scan per subject using intraclass correlation analysis as established Amico et al. (Amico and Goñi, 2018). Coefficients derived from ICC are widely used as a reliability index in test-retest analyses, reflecting the percentage agreement between two units of measurement (e.g. an edge) within the same group

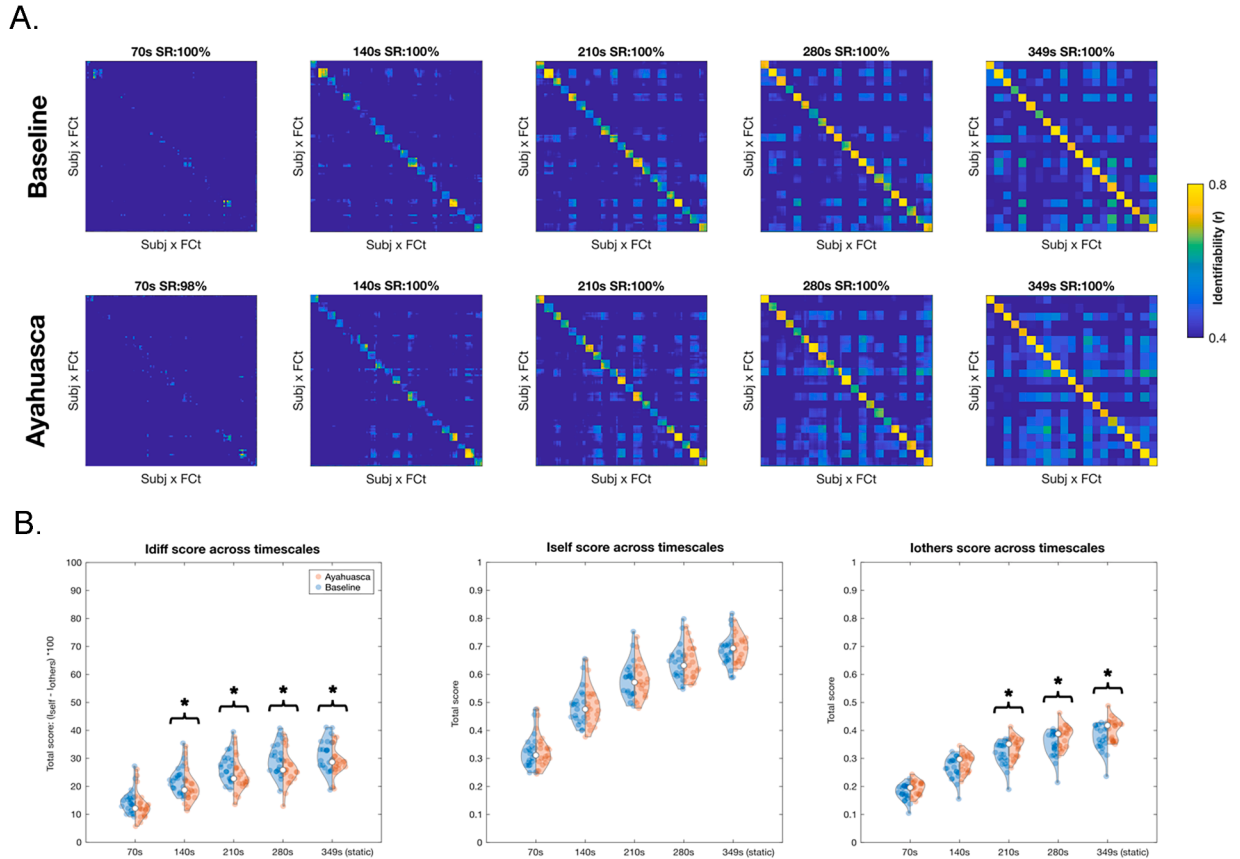


Fig. 4. Whole-brain measures of dynamic identifiability. (A) Dynamic identifiability matrices at five different window lengths (70, 140, 210, 280 and 349 s) for each condition. The dynamic differential identifiability (I_{diff}) values and success rates (SRs) on top of each matrix provide two complementary scores of the fingerprint level of the dataset across temporal scales (see Methods). (B) Violin plots highlighting differences in each identifiability metric (I_{diff} , I_{self} , I_{others}) per timescale. Subjects are represented with single points. Two-tail significance is denoted as follows: $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

(e.g. a subject) (McGraw and Wong, 1996; Noble et al., 2019). The greater the ICC value, the greater the consistency these two units hold. For reference, ICC values below 0.40 are suggested to be poor/unreliable whereas those beyond 0.90: excellent/congruent (Cicchetti and Sparrow, 1981).

We employed this approach under the assumption that subsets of highly stable edges (edges holding high ICC scores) across test-retest sets edges are major drivers of each state's connectome fingerprint. For a FC, this generates a square symmetrical ICC matrix of size N^2 , where N is the number of brain regions (see Figs. 4 and 5) for a specific timeframe. From this, nodal ICC strength (i.e. regional ICC scores) can be characterised by summing ICC values across rows. We also extrapolated network identifiability by averaging ICC values of within and between network edges, producing 7×7 ICC fingerprint matrices corresponding to our Yeo parcellation. Note that edges were thresholded according to lower bounds of ICC (0.40).

In the case of dynamics, there might be FC frames where identification is higher than others. Consequently, this might not reflect the average behaviour depicted by dI_{diff} thereby skewing ICC estimates. To cover this necessity, for each subject session, we sorted the dFC frames in test-retest according to their similarity, from highest to lowest, based on their dI_{self} value. We then recalculated dI_{self} , dI_{others} , and dI_{diff} when iteratively adding dFC frames one at the time, starting from the best matching ones and then proceeding based on their similarity values in order to end with a "top" frame for each timescale on which our ICC analyses could be performed. As a supplementary analysis we also

examined the relationship between FC variability and stability by calculating the standard deviation of functional connectivity for each frame (see figure S4.).

With the expectation that subsets of static edges might primarily contribute to each condition's identifiability, we sorted these according to their thresholded ICC values computed on the baseline condition (baseline). Edges were added in a descending fashion, with I_{diff} being recalculated at each iteration of 50 edges. By defining I_{diff} as a target variable, the optimization problem during edge ranking of differential identifiability (total uniqueness) is then simplified to maximizing I_{diff} . We selected the sober condition as an index to visualise the evolution of normative drivers maximally contributing to I_{diff} (in other words, edges driving a subject's total fingerprint score).

2.8. Edgewise prediction of subjective experience

In light of the individual nature of subjective experience and connectome fingerprints we opted for an iterative multilinear regression modelling approach (MLR) similar to connectome predictive modelling (Shen et al., 2017). Aiming to assess the relative explanatory power of connectome fingerprints for inter-individual differences in subjective experience, we employed principal component analysis (PCA) decomposition (Jolliffe and Cadima, 2016) of the functional connectivity values of highly identifiable edges (subjects $\times$ edges). An unsupervised exploratory approach, PCA is a dimensionality reduction approach used to transform a high-dimensional dataset into a lower-dimensional space

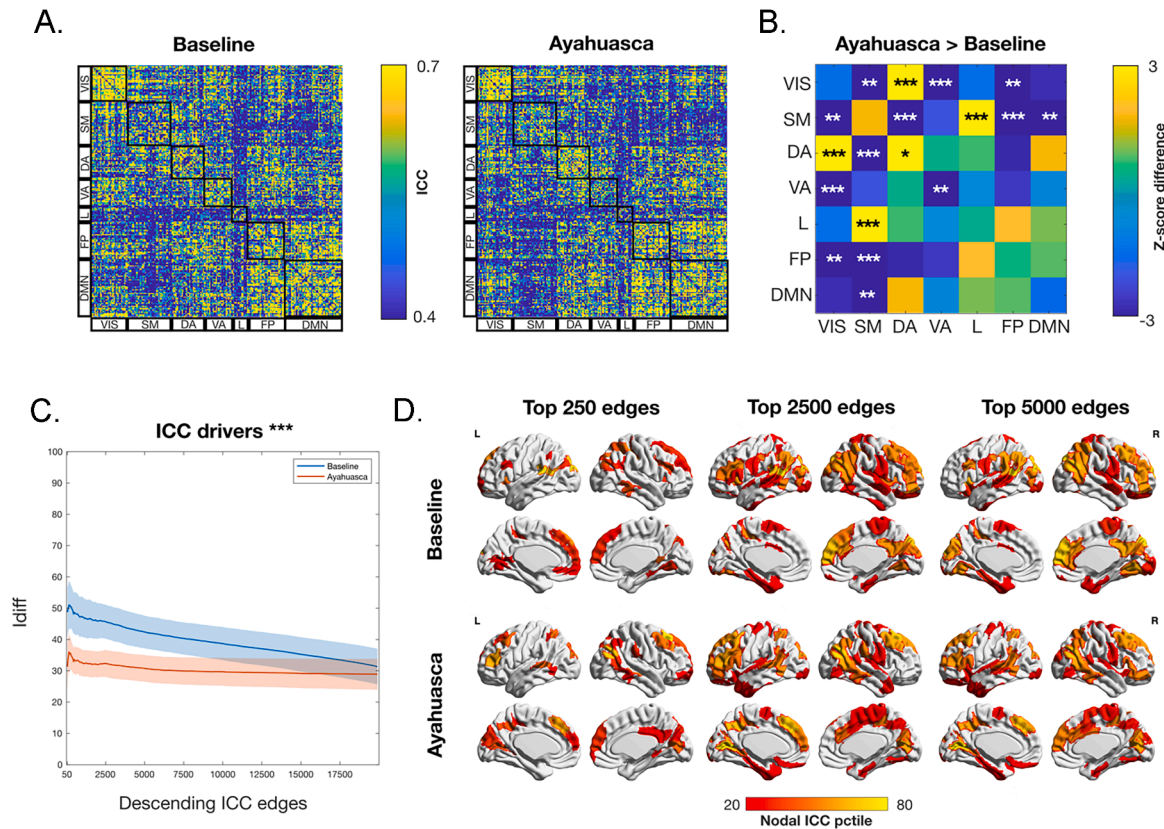


Fig. 5. Spatial specificity of connectome fingerprints. (A) Edgewise intra-class correlation (ICC) matrices per condition (baseline, ayahuasca). The ICC matrices are shown thresholded at 0.4. All 7 functional networks as defined by Yeo et al. (see Methods) are highlighted by black boxes: VIS = visual network; SM = somatomotor network; DA = dorsal attentional network; VA = ventral attentional network; L = limbic network; FPN = fronto-parietal network; DMN = default-mode network. (B) Differences in network ICC values between conditions. For each condition, ICC edgewise scores are grouped per Yeo functional network and compared using Bonferroni-corrected two-tail sign-rank testing. Approximated z-scores are then extrapolated and plotted for ease of visualisation. (C) Identification of top fingerprinting edges. I_{diff} scores were obtained by iteratively calculating identifiability matrices for each condition, ranked according to those contributing the most to baseline identifiability (as per ICC values). Lines represent condition means, with shading reflects the standard deviation of I_{diff} across subjects at each step. Nodal strength (sum across unthresholded ICC regional matrix rows) across subsets of top fingerprinting edges per condition. For each render percentiles are shown (from 20th to 80th percentile). For all plots, two-tail significance is denoted as follows: $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

while retaining most of the sample's variance. Notably, the resultant principal components (PCs) are a set of uncorrelated vectors that represent the directions of maximum variance in the data. Consequently, this allows us to isolate a subset of maximally heterogeneous and independent FC motifs based on idiosyncrasy which may best reflect variable behavioural traits. This decomposition was applied in an iterative fashion for model selection. Firstly, all static edges driving idiosyncrasy in the ayahuasca condition were sorted in descending order according to their ICC value. At each iteration of 50 edges, we performed a PCA decomposition of these demeaned n , retaining three PCA components ranked according to explained variance. Then, a MLR was built for each subjective effect measure, comprising these three PCA components as predictors of interest alongside two covariates: singing (self-reported internal singing during the resting-state) and scrubbing (number of valid volumes). Absence of multicollinearity was assessed using variance inflation factor (VIF) (Craney and Surles, 2002). At each iteration, we strengthened the reliability of our model using k -fold cross-validation (Fushiki, 2011) with $k = 5$. Specifically, k iterations were performed and at each iteration the k^{th} subgroup was used as a test set. For each iteration, the Spearman's correlation coefficient between predicted and actual inventory values was calculated and considered as a performance score. We assessed the reliability of this performance score against surrogate models, computed using a set of randomly permuted edges at each step. For each variable of interest this process was repeated 100 times. This iterative permutation approach provided a cut-off point for edges maximally contributing to predictive performance by identifying a set number of edges at which model performance outperformed its surrogate model's performance standard deviation while balancing explained variance (see Figure S6.). It is worth mentioning that while it is common practice to define the number of retained PCs (1... p) such that their cumulative explained variance is close to 100%, our decision of retaining 3 PCs was defined by a I) Tolle et al. demonstrating $p = 3$ provides an optimum trade-off between explained variance and overfitting in idiosyncrasy-informed modelling (Tolle et al., 2023) and II) using an unrestricted approach would have required us to change p for each iteration and null models, limiting model comparisons.

2.9. Statistics

Statistical analyses were carried out in MATLAB 2019b. Shapiro-Wilks was firstly used to assess the normality of all measures. Control variables (subjective effects, PK) were assessed by means of one-tailed t -tests against zero. All other outcome measures were analysed in a two-tailed fashion according to their normality; either by Wilcoxon sign-rank (W) or paired-sample t -tests (t) with Cohen's d effect sizes (d) being provided for each. Observed static identifiability scores (true) values were examined against corresponding null-distributions following a permutation testing framework (see supplementary). Regarding network-based statistics, we retrieved a conservative Bonferroni-holm corrected p -value (p_{bonf}) according to the number of unique elements in each matrix. The alpha criterion of significance for all inferences was set at $p < 0.05$.

3. Results

Experienced members of Santo Daime were enrolled in a fixed-order, within-subject, observational study. A baseline (sober) resting-state fMRI was followed 1 day later with a second acute (ayahuasca) fMRI scan 90 min after communal intake (i.e., peak effects). The study also entailed pharmacokinetic sampling, questionnaires pertaining to retrospective drug effects and aspects of "work" during resting-state (see Methods). Of the 24 patients recruited, 3 were excluded due to excessive fMRI head motion. Demographic information pertaining to the imaging sample can be found in Table S1.

3.1. Acute effects of ayahuasca

Ayahuasca intake was associated with increased ratings on all (sub) dimensions of the 5D-ASC (max. oceanic boundlessness: $t = 7.29$, $p < 0.0001$, $d = 1.59$; min.changed meaning of percepts: $t = 3.90$, $p < 0.0004$, $d = 0.85$), and for the EDI ($t = 7.15$, $p < 0.0001$, $d = 1.56$). Mean ratings of oceanic boundlessness (OBE. 33.1% [SEM: 4.54]) and visual restructuralization (VR. 26.24% [SEM: 4.37]) were most affected. Tandem pharmacokinetic analyses also demonstrated serum concentrations of DMT (the principal psychoactive constituent of ayahuasca) were significantly greater than zero at both 60 ($t = 4.82$, $p < 0.0001$, $d = 1.14$) and 160 min ($t = 5.16$, $p < 0.0001$, $d = 1.18$) after intake.

During resting-state acquisition, participants reported significantly more internal singing under ayahuasca ($W = 58$, $p = 0.0261$, $d = 0.63$). Levels of engagement in meditation did not significantly differ between conditions. A full characterisation of all inventories and serum alkaloids can be found in the supplementary materials.

3.2. Quantifying whole-brain fingerprints

Connectome fingerprinting provides a window into the "uniqueness" of one's functional connectivity (Amico and Goñi, 2018; Van De Ville et al., 2021; Sorrentino et al., 2021). This approach stems from the simple assumption that a FC should hold greater similarity between test-retest scans of the same subject than between different subjects (Finn et al., 2015). By computing an "Identifiability matrix" we can extrapolate for a subject whole-brain metrics reflecting both the *intra*-individual (I_{self}) and *inter*-individual (I_{others}) variability of their functional connectome (see Methods, Fig. 2). These measures can be compounded as an overall fingerprinting score (I_{diff}) in other words, how well a subject can be identified within a group of other subjects based on their connectome. As a first pass, we explored changes to whole-brain fingerprints and their dynamical counterparts. We derived measures of identifiability for static FCs (Fig. 3) and their dynamic equivalents (Fig. 4) by replicating our analyses across increasing window lengths (see methods). In each case we also provide success rate (SR) (Finn et al., 2015; E. Sareen et al., 2021) as a supplementary assessment

Static identifiability. As depicted in Fig. 3B, sign-rank testing revealed the differential identifiability (I_{diff}) of each participant was significantly diminished under ayahuasca ($W = 53$, $p = 0.0298$, $d = 0.35$) reflecting an overall reduction in a subject's FC idiosyncrasy. If we examine its constituents, this effect was driven by a significantly increased I_{others} score ($t = 2.72$, $p = 0.0131$, $d = 0.59$). In other words, participant connectomes significantly mirrored one another's under ayahuasca, depicted by the saturation of off-diagonal elements under ayahuasca (Fig. 3A). Remarkably, subjects continued to retain high I_{self} ($p > 0.05$) and SR scores, reflecting a preserved idiosyncrasy under ayahuasca. It should be noted that each condition's I_{diff} and SR scores were also significantly greater than their null equivalents following permutation testing ($p = 0.001$, see Figure.S1).

We then examined how dissimilar might the constituent edges of a subject's fingerprint scores be under ayahuasca. Hybrid equivalents of our identifiability matrices (see Fig. 2, Methods) enabled us to derive the "distance" of each subject's score (ie their relative composition) from baseline. Doing so, we identified a greater dissimilarity between a subject's functional connectome under ayahuasca versus baseline, with both $I_{\text{selfHybrid}}$ ($t = -8.67$, $p < 0.0001$, $d = 1.89$) and $I_{\text{diffHybrid}}$ ($t = -7.94$, $p < 0.0001$, $d = 1.74$) significantly reduced. In other words, while a subject's scalar self-identifiability score remains the same under ayahuasca, its constituent edges differ relative to baseline. This can be visualised as the faded diagonals in the off block "Hybrid" elements of Fig. 3A and further reflected by a low I_{clinical} score (39.21%).

Dynamic identifiability. Might specific timescales of neural processing further account for these global differences? Repeating this previous analysis across increasing window size, reveals an equivalent pattern. As per prior work (Van De Ville et al., 2021), dynamic (I_{diff})

increased steadily with longer window lengths (Fig. 4A) as a by-product of the increasing number of timepoints for dFC computation with early dynamical fingerprints (designated by clear diagonal elements) arising at shorter temporal intervals.

Replicating our analyses across each temporal scale, we observed the equivalent patterns of change. As shown in Fig. 4B, I_{diff} was significantly reduced under ayahuasca in a temporally selective manner (across 140s–349 s; max.220s: $W = 51$, $p = 0.0250$, $d = 0.41$). This effect was partly accounted by I_{others} increasing across select frames (210–349 s max.349s: $t(20) = 2.72$, $p = 0.0131$, $d = 0.59$). Once more, I_{self} was found to remain stable at all timescales ($p > 0.05$). For all windows, $I_{selfHybrid}$ was significantly reduced under ayahuasca (max.210s: $t = 9.39$ $p < 0.0001$, $d = 1.74$), as well as $I_{diffHybrid}$ (max.140s: $t = 9.81$ $p < 0.0001$, $d = 2.16$).

3.3. Select edges mediate reductions in connectome identifiability

Identifying global changes to each subject's connectome fingerprint under ayahuasca, we then sought to understand their spatiotemporal profiles. To do so, we applied an edgewise ICC to investigate the fingerprinting value of edges (see Methods, Fig. 2) pertaining to canonical resting-state networks (RSNs).

Static connectomes. We observed global reductions in ICC scores ($W = 92,864,811$, $p < 0.0001$, $d = 0.05$) under ayahuasca. While this suggests a connectome-wide drop of temporal stability, individual RSNs have varying levels of importance for fingerprints and may be differentially affected (Fig. 5A). Focusing on network properties (Fig. 5B), within-network analyses following Bonferroni correction revealed significant reductions in stability for the ventral attentional (VA. $W = 8414$, $p_{bonf} = 0.0014$, $d = 0.27$) and converse increases for the dorsal attentional network (DA. $W = 30,021$, $p_{bonf} = 0.0283$, $d = 0.20$) under

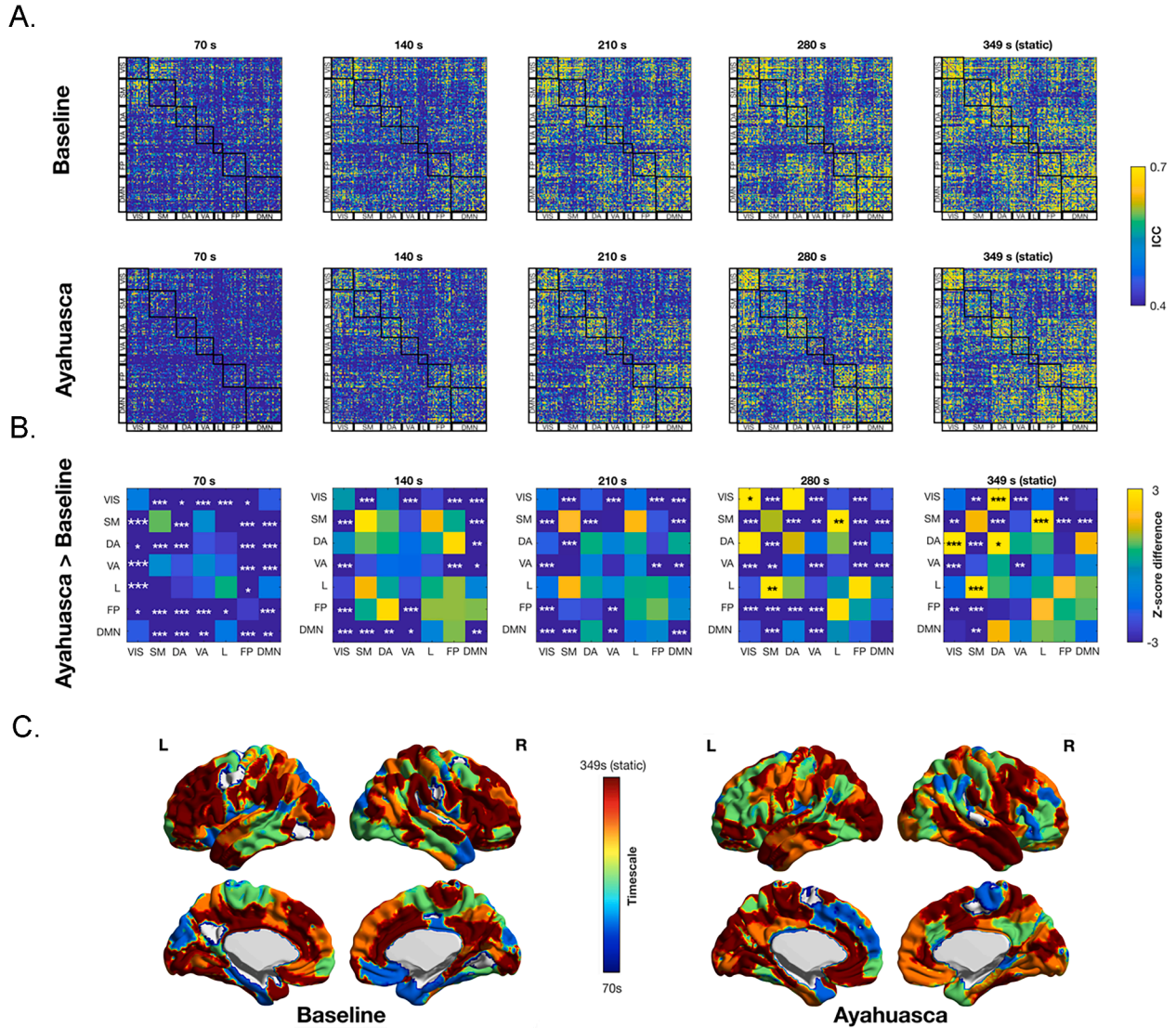


Fig. 6. Temporal specificity of connectome fingerprints. (A) Mean edgewise intra-class correlation (ICC) matrices per condition (baseline, ayahuasca) at each timescale. The ICC matrices are shown thresholded at 0.4, the cut-off for a reliable ICC score (Finn, 2021). All 7 functional networks as defined by Yeo et al. (see Methods) are highlighted by the black boxes: VIS = visual network; SM = somatomotor network; DA = dorsal-attention network; VA = ventral-attention network; L = limbic network; FPN = fronto-parietal network; DMN = default-mode network. (B) Differences in network ICC values across timescales. For each condition and per window, ICC edgewise scores are averaged across Yeo functional networks and compared using Bonferroni-corrected two-tail sign-rank testing. Approximated z-scores are then extrapolated and plotted for ease of visualisation. Lighter hues reflect increases in ICC values under ayahuasca whereas darker ones reflect diminishments under ayahuasca. (C) Mean temporal peaks of nodal stability. Maximum values across temporal profiles at each brain node are overlaid onto a brain render to map the time scales of human brain fingerprints. The maximum value for each brain node was derived from ICC nodal strength values (sum across ICC regional matrix rows) at each window per condition. For all plots, two-tail significance is denoted as follows: $p < 0.05^*$, $p < 0.01^{**}$, $p < 0.001^{***}$.

ayahuasca. In contrast, inspection of between-network pairs reveals reductions in stability were primarily attributable to the visual (VIS) and VA functional subsystems ($W = 36,803$, $p_{\text{bonf}} < 0.0001$, $d = 0.49$). Within these, certain network pairs such as SM-L ($W = 25,535$, $p_{\text{bonf}} < 0.0001$, $d = 0.26$) and VIS-DA connectivity ($W = 122,508$, $p_{\text{bonf}} = 0.0008$, $d = 0.14$) exhibited greater stability under ayahuasca.

Given that subsets of highly synchronous edges are important contributors to normative connectome fingerprints (Amico and Goñi, 2018; E. Sareen et al., 2021; Van De Ville et al., 2021; Sorrentino et al., 2021), we investigated how they might shift in importance under ayahuasca. Ranking baseline edges from most to least stable, we recalculated each subject's identifiability 50 edges at a time. Fig. 5C shows that, while baseline, or "normative" identifiability can be maximised within 250 edges, the contribution of these edges to fingerprinting drops markedly under ayahuasca ($\text{Idiff. } t(249) = -10.12$, $p < 0.0001$, $d = 2.38$). Therefore, edges otherwise normally "driving" a subject's identifiability are no longer significant contributors when under the influence. Rather, a reconstitution of edge importance becomes apparent when examining their nodal equivalents (Fig. 5D). One can notice connections implicated in hubs pertinent to DA, VA and SM networks are instead primarily replaced by those pertinent to the DMN.

Dynamic connectomes. We then examined differences in spatial ICC patterns as a function of time by repeating our analysis across each timescale. As window length increases, one can note different networks appearing at different rates, such as sensory networks at shorter intervals or the DMN at slower scales (Fig. 6A). This gradient highlights the varying temporal prerequisites of RSN fingerprints (Van De Ville et al., 2021). Our ICC analyses revealed global reductions in dFC stability across all measured timescales (max.70s: $W = 79,109,036$, $p < 0.0001$, $d = 0.18$) under ayahuasca.

As shown in Fig. 6B, network-based analyses revealed diffuse changes to the stability of dynamic functional connectivity under ayahuasca (for a full characterisation see tables S5.1–5). Novel reductions in within-network edge stability were identified across increasing windows of time for: the DMN (max.210s: $W = 171,684$, $p_{\text{bonf}} < 0.0001$, $d = 0.18$); VA (max.280s: $W = 6549$, $p_{\text{bonf}} < 0.0001$, $d = 0.37$) and DA networks (max.70s: $W = 8250$, $p_{\text{bonf}} = 0.0003$, $d = 0.245$). Contrarily, VIS network edges exhibited greater stability at 280 s ($W = 48,931$, $p_{\text{bonf}} = 0.0122$, $d = 0.13$). In parallel, reductions in between-network edge stability populated all scales. This attenuation could be primarily ascribed to edges involved in between-network SM and VIS connectivity (max.VIS-SM (70 s): $W = 82,429$, $p_{\text{bonf}} < 0.0001$, $d = 0.47$). Furthermore, previously identified static increases in between-network SM-L connectivity stability was found to be time-dependant (280 s. $W = 18,936$, $p_{\text{bonf}} < 0.0001$, $d = 0.16$). We also examined whether changes to standard deviation of different RSNs (see supplementary materials) might also help explain changes to the topography of edge stability. In this regard, while between-network reductions in functional connectivity variability was observed, no clear association could be ascertained (see Figure.S4).

We next asked whether altered fingerprint dynamics under ayahuasca could also be reflected at a regional level of brain organisation. Identifying each region's ICC maximum, we summarised their temporal optimums as a brain render (Fig. 6C). Typically, transmodal regions comprising association cortices, "peak" at longer temporal windows whereas unimodal regions, such as primary sensory areas, arise early on (Van De Ville et al., 2021). While this was the case at baseline, this temporal gradient shows an inversion effect following intake, best demonstrated by regions such as the prefrontal cortex peaking early on or vice versa for unimodal areas such as the visual cortex.

3.4. Connectome fingerprints are predictive of perceptual drug effects

We lastly performed an exploratory analysis investigating the behavioural relevance of connectome fingerprints. We hypothesised that

highly idiosyncratic edges under ayahuasca could also predict meaningful aspects of a subject's subjective experience. To assess the behavioural relevance of idiosyncratic edge connectivity, we built an iterative multilinear model approach comprising PCA components of subsets of ICC-ranked edges as predictors of interest for our subjective effect measures. A k -fold cross validation revealed that peak predictive performance for the 5D-ASC dimensions Visual Reorganisation (VR) and Auditory Alterations (AA) was achieved using the top 3000 most stable edges (see Figure S6.), with predictive performance for all other outcome measures being no better than the null model.

Other than three edge-based PCA components, each model also comprised two other predictors: scrubbing and internal singing. We found that of all PCA components, only PCA3 significantly increased the predictive power of the model for VR ($F(4,20) = 1.98$; $R^2 = 0.45$; $p = 0.0396$; $\beta = -5.59$) and AA ($F(4,20) = 4.8$; $R^2 = 0.70$; $p = 0.0016$; $\beta = -2.17$). Together, this finding might reflect the fact that subsets of edges are predictive of each dimension only when considering the idiosyncrasy of functional connectivity. Edges most implicated were primarily found in FPN, DMN and DA hubs as well as the regions pertaining to the VIS (Fig. 7C). As a precaution we also examined whether the motion (nScrub) or shared behaviour (Singing) were relevant predictors, finding no significant contribution to model performance.

3.5. Additional control analyses

For completeness, we performed a series of quality control analyses on our primary identifiability findings. Specifically, we (i) repeated our main analyses using a coarser Schaefer 100-node parcellation, (ii) using censored fMRI timeseries, (iii) assessed differences in motion metrics between conditions, (iii) examined split-half differences in primary motion outcomes per scan, (iv) evaluated their association with all sFC and dFC identifiability outcomes. Our findings appear robust to motion, parcellation resolution and replicable across different denoising strategies (Figure.S2-S5.).

4. Discussion

Here, we leveraged the understanding that an individual's brain functional connectivity profile is both unique and reliable to document how the inherent features of a subject's functional connectome might transition into a collective altered state of consciousness. Using the concept of connectome fingerprinting outlined by Amico et al. (Amico and Goñi, 2018), in a cohort of 21 Santo Daime members taking part in the ritualistic use of ayahuasca, we were able to detect for each subject a significantly greater proportion of shared functional connectivity traits across different timescales of neural processing. Furthermore, we show that this shared variance is accompanied by the reconfiguration of keypoint edges pertinent to higher-order functional subsystems, otherwise driving normative brain "fingerprints". Equally, we show that the instability of edges is likely relevant to experiential differences given that they can be used to predict aspects of an individual's subjective experience. All in all, these findings point to the general blueprint of a subject's inherent resting-state functional connectivity shifting to overlap with fellow members (see Fig. 8 for an analogy). Ultimately, subject-level approaches such as those presented herein highlight the potential to discover personalised fMRI-based connectivity markers that may eventually be used to trace a subject's functional connectome across states of consciousness.

4.1. The collective use of ayahuasca is associated with shared connectome fingerprints

Whereas we found the I_{diff} of a subject's connectome was diminished under ayahuasca, this reduction could be ascribed to the increased contribution of I_{others} . In other words, individual connectome fingerprints were found to be less idiosyncratic under the influence of

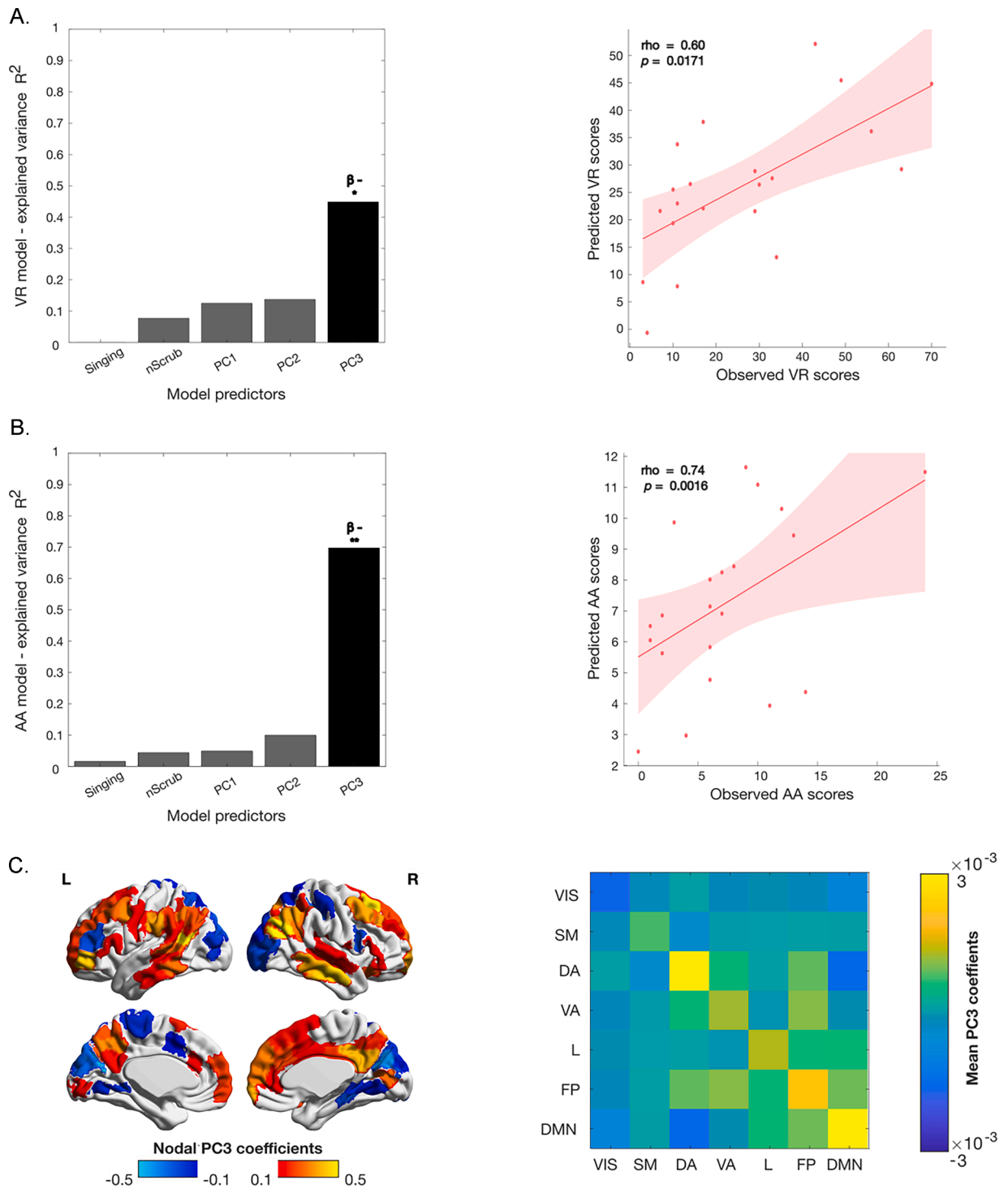


Fig. 7. Peak predictive multi-linear model of subjective effects. Following an ICC derived feature selection comprising k-fold validation and null-modelling (see Methods), 3000 edges were found to yield explanatory power. **(A)** Visual restructuring (VR) peak performance model. Left: The additive linear model consists of two nuisance variables (nScrub, Singing), and three PCA predictors (PC1–3); Right: Scatter plot of the observed VR scores versus predicted model VR scores. **(B)** Auditory alteration (AA) peak performance model. As before, left: incorporated predictors in an additive linear model, Right: observed vs predicted AA scores. **(C)** PC3 coefficient loadings. Nodal strength (sum across regional matrix rows) and corresponding network means of the significant PC3 coefficient loadings for the top 3000 edges are depicted. For all panels, significant predictors are denoted as follows: $p < 0.05^*$, $p < 0.01^{**}$ with β^- indicating that its beta coefficient is negative.

ayahuasca. When factoring in the practices of Santo Daime, this could echo the shared practices carried out by church members. Recent replication of the present methodology in a sample of recreational users not holding shared rituals yielded compatible *inverse* findings of enhanced FC idiosyncrasy (reductions in I_{others} and elevations in I_{diff}) under the 5-HT_{2A} agonist psilocybin (Tolle et al., 2023). Evidence from

classical psychedelics suggests an “unconstrained” state of cognition of few deliberate or automatic constraints, featuring a large amount of hyper-associative thinking and diminished reality-testing (Girn et al., 2020). While ayahuasca experiences are highly subjective, followers of Santo Daime share stereotyped behaviours otherwise absent in recreational users, such as singing or attentional deployment, synonymous

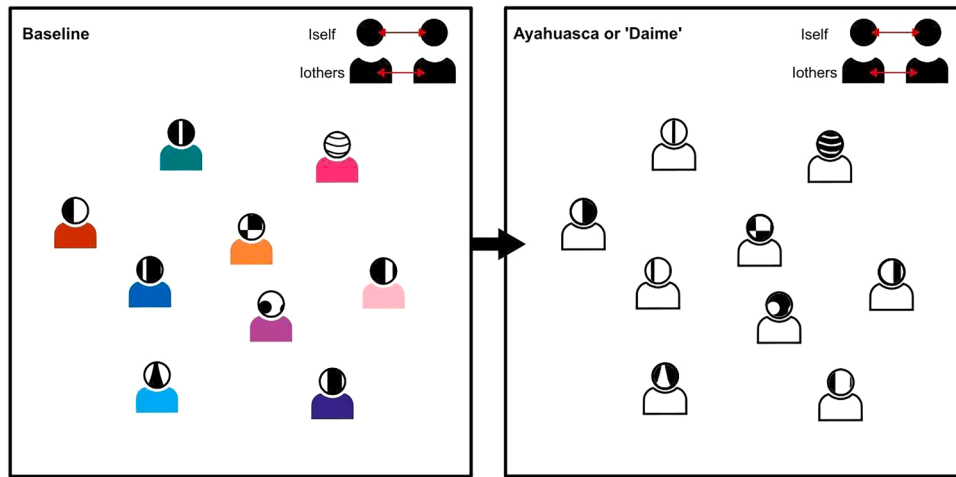


Fig. 8. Graphical analogy of connectome fingerprint shifts under ayahuasca. In everyday life people rarely dress one and the same. Ordinarily, an individual might choose their attire based on personal preference, such as a colourful shirt. The colour palette that we might choose would represent our distinctiveness, in turn differentiating us from others (I_{others}). Throughout our day, I_{others} might vary, given others with distinct preferences might come and go. Now say in a different scenario, such as a Santo Daime ceremony, we were to abide by the dress code of a white uniform. Even if others might come and go, our similarity to others (I_{others}) at the ceremony would be high since the colour white is mandated throughout the event for all participants. However, if the uniform were to be contrasted to daily life ($I_{\text{othersHybrid}}$) we might find it to be just as dissimilar as any other coloured shirt that we might come across on an average day. In parallel the constituents of our self-identity (I_{self}) may equally be denoted as a unique pattern; Whereas the total level of distinctiveness is unlikely to change regardless of circumstance, an acute perturbation by a pharmacological agent such as ayahuasca may change the pattern's constitution ($I_{\text{selfHybrid}}$).

with a constrained state of cognition. For example, prior imaging work with church-goers has previously drawn parallels with the induction of a task-active state, exemplified by suppressions in DMN activity (Palha-no-Fontes et al., 2015) normally associated with external goal-directed attention such as task engagement or focused attention meditation (Scheibner et al., 2017; Tripathi and Garg, 2022). Furthermore, studies with normative samples demonstrate that the inter-individual variability of a sample is diminished when engaging in a task battery, proportionally to cognitive load (Geerligs et al., 2015; Finn et al., 2017). Thus, an interpretation could be that such a constrained mental state is disseminated across individual connectivity matrices.

While the appearance of shared functional connectome fingerprints pertained to dFC timeframes at which complex cognition emerges (Van De Ville et al., 2021), it however, cannot be definitively stated whether this shared variance is solely attributable to group behaviour. Ayahuasca itself was not administered in any other context, such as for example, the individuals alone. It may very well be the case that ayahuasca alone may propagate across connectivity matrices in the manner described, given DMT's particularly potent effects on functional brain organisation (Timmermann et al., 2023). Without future work disentangling the many spontaneous cognitive processes arising from resting-state functional connectivity, the synergistic influence of cognitive state is uncertain. More so than tasks, “ground truth” approaches for pharmaco-imaging such as films (Meer et al., 2020) or other integrated designs (Finn, 2021) paired with subject-level dynamical analysis approaches (Shafiei et al., 2020) may hold promise in tagging the behavioural relevance of dFCs for attention-impairing drugs such as these.

4.2. Constituents of connectome self-identity are mutable under ayahuasca

Studies have repeatedly demonstrated the remarkable consistency of inherent functional connectivity patterns across participants and mental states (Gratton et al., 2018; Cole et al., 2014). Most of a subject's uniqueness or inter-session variance (between 63 and 87%) can be explained by commonalities in functional connectivity architecture between states (Geerligs et al., 2015). Here, we show that contrary to a phenomenological loss of self-identity, the degree of self-identifying

connectivity (I_{self}) is also preserved under ayahuasca. Psychedelics are described to produce a wide-scale discoordination of brain activity (K. H. Preller et al., 2018; Madsen et al., 2021) denoted by a structural-functional uncoupling (Luppi et al., 2021). To take the view that fingerprinting comprises fixed anatomical loci which assimilate several information sources to plan coherent behavioural responses (Avena-Koenigsberger et al., 2018) then their impaired integration as observed under psychedelics should also lead to a diminished I_{self} . However, connectome fingerprinting of clinical populations exhibiting structural-functional uncoupling show no differences in I_{self} against healthy controls (Stampacchia et al., 2022). Instead, it is now known the total blueprint of functional connectivity does not constitute discrete networks but is rather best described by more mutable local and global gradients (Margulies and Smallwood, 2017) which are likely susceptible to pharmacological perturbation. Indeed, our finding of diminished $I_{\text{selfHybrid}}$ may instead reflect a general functional reconfiguration of inherent signalling traits, synonymous with the apparition of a novel functional connectivity architecture.

4.3. Local shifts in functional connectivity stability drive altered connectome fingerprints

At a fundamental level, we also looked at the edgewise contributors to identification which might explain the observed reconfiguration of identifiability. Previous work has closely implicated temporal stability (as defined by ICC) of regional functional connectivity as driving connectome fingerprints (Van De Ville et al., 2021). Using ICC, we observed global reductions in edge stability at all measured timescales under ayahuasca, mirroring the patterns of functional change under psychedelics. 5-HT_{2A} agonists have been found to produce brain-wide increases in signal complexity (Varley et al., 2020; Viol et al., 2017), which may consequently limit the temporal concordance of edge pairs. Before continuing, it should be noted that functional connectivity and ICC are not interchangeable but rather complementary methods (Noble et al., 2021) and future work should continue to examine the mediating relationship between the differing measures of complexity available, ICC and connectome identifiability (Liu et al., 2020).

Given that certain edges drive a subject's normative fingerprint, we examined how regional contributions to identifiability evolve under

ayahuasca. While a subset of 250 edges could maximally define a subject's fingerprint at baseline, their importance markedly dropped under ayahuasca. Disseminated across higher-order association cortices, these regions are shown to encode the majority of inter-individual variance (Finn et al., 2015; Mantwill et al., 2022). Importantly, it has been previously hypothesised that the appearance of a desegregated functional architecture under psychedelics stems from the impairment of these same functional subsystems (M. K. Doss et al., 2021). However, frontotemporal DMN nodes central to the effects of hallucinogens (Carhart-Harris et al., 2014) emerged as the focal point for a subject's identifiability under the influence, expressing greater stability. The DMN has been implicated in different aspects of conscious experience, such as ongoing cognition (Smallwood et al., 2021), spontaneous thought (Kam et al., 2022), rumination (Chen et al., 2020), and self-referential processing (Lebedev et al., 2016) and its select prevalence may further highlight a diminished variability of subjective experience (Zamani et al., 2022).

These shifts in stability were also pronounced at a network level. While functional networks do not equally contribute to an individual's fingerprint, each functional subsystem is thought to have temporal "peaks" in stability (Van De Ville et al., 2021; Mantwill et al., 2022). Here, links pertaining to VIS and SM between-network connectivity exhibited greater instability under ayahuasca, across all examined temporal windows. These findings appear in line with patterns of variable connectivity in sensory networks that have been observed under psychedelics (Preller et al., 2020; K. H. Preller et al., 2018), thought to reflect a dedifferentiation of hierarchical organisation (Girn et al., 2022). Furthermore, clustering-based dFC approaches with LSD and psilocybin have shown an increased fractional occurrence and dwell time of alternating states of hyperconnectivity (Singleton et al., 2022; Olsen et al., 2022; Lord et al., 2019) which may account for the stochasticity of edge stability at each temporal window. It could be therefore suggested that the outcome of a dedifferentiation of functional hierarchies under psychedelics may also extend to their temporal organisation. Suggesting this, both ends of the continuum of hierarchical organisation became temporally more similar to one another under ayahuasca; with unimodal regions (eg. parietal operculum, visual cortex) associated with rapid multisensory processing now peaking in stability at longer timescales and transmodal regions (eg. prefrontal cortex, posterior cingulate cortex) otherwise exhibiting longer, integrative firing patterns maximal at shorter timescales. Applying approaches to examine the temporal propagation and latency of brain activity (Raut et al., 2019; Mitra et al., 2014) under psychedelics may highlight new explanations for their effects on network architecture.

4.4. Connectome fingerprints are relevant to the subjective ayahuasca experience

Assuming ayahuasca experiences are highly individual, might subject-level shifts in functional connectivity also help predict overlaying subjective experiences? To explore this hypothesis, we devised a data driven PCA approach to assess the behavioural relevance of highly identifiable edges. Our results suggest that subsets of highly stable edges not only drive a subject's identifiability under the influence but also hold explanatory power for the AUD and VIS dimensions of the 5D-ASC. In practice, by decomposing the total variance of a FC signal to a reduced number of uncorrelated components, PCA offers the opportunity to isolate maximally behaviourally relevant FC motifs in a subset of highly explanatory components. If group differences were highly explanatory of experiential scores, then a single component (PC1) capturing most of the sample variance might have emerged as a principal model contributor. Instead, our results were contingent on the inclusion of PC3 as a predictor of interest, suggesting higher-order PC deviations capturing the sample heterogeneity in FC were most relevant to the visual and auditory effects of ayahuasca. Furthermore, predictive edges were found to span primarily both higher-order systems (e.g., DMN) and primary

systems (e.g., VIS), with the former contributing strongly to behavioural prediction as per prior work (Finn et al., 2015; da Silva Castanheira et al., 2021; Mantwill et al., 2022). Given their developmentally late maturation (Xia et al., 2022), susceptibility to individual environmental effects (Valk et al., 2022), dense 5-HT_{2A} expression (Beliveau et al., 2017) and coordination of multisensory integration in comparison to primary systems (Margulies and Smallwood, 2017), higher-order regions may more easily account for divergent phenomena, more so than primary systems, themselves partially influenced by the temporary states of each individual during scanning (Agcaoglu et al., 2019). While these proof-of-principle analyses are concordant with more recent findings of behaviourally relevant connectome fingerprints under psilocybin (Tolle et al., 2023), deriving reliable phenotypes from resting state measures is contingent on large sample sizes (Liu et al., 2023). Thus, replicating the present findings with an exhaustive PC search space in a larger sample size will serve to not only define the reliability but also the validity of these FC motifs as markers of subjective experience.

4.5. Limitations

The present work comes with several limitations. Importantly, members of Santo Daime are not reflective of the general population. Drinking ayahuasca several times a month, members likely exhibit a level of habituation to the drug's effect. Furthermore, 5-HT_{2A} agonists are potent psychoplastogens (De Vos et al., 2021) likely inducing structural alterations after prolonged use. For example, cortical thickness analyses of Santo Daime members have demonstrated an association between significant thinning in midline structures and self-transcendent personality traits (Bouso et al., 2015). There is an abundance of studies showing how between-subject differences in white matter integrity are intimately related to interindividual variability in functional dynamics (Genon et al., 2022; Gu et al., 2021) and likely identifiability (Takao et al., 2015). As with observational studies, these findings are subject to confounding effects regarding dosage, blinding, sample inclusion criteria and expectancy. Adequate blinding in studies comprising experienced users continues to be an unresolved factor in the field, due to a subject's immediate recognition of a drug's effect (or non-effect). This is further accentuated here, given the ritual elements. Future studies with a diverse sample of ceremonial users may benefit from not only counterbalancing and adequately blinding the agent in question, but also surrounding practices mediating experiential outcomes. With regard to the methodology, it is well-known that head motion due to its potential for skewing functional connectivity estimates (Power et al., 2014; Power et al., 2012) is likely a confounder in the study of inter-individual differences (Finn et al., 2015). If treated as a "state" characteristic for subjects, joint differences amongst members of a group might account for a proportion of between-subject variability. Whereas numerous steps were taken to exclude its influence, it is unknown to what degree factors such as motion, respiratory fluctuations or arousal level may prevail in shorter dFC windows. Future studies should aim to replicate this workflow using framewise approaches such as dynamic conditional correlations (Lindquist et al., 2014) or phase coherence estimation (Honari et al., 2021). Per prior work (Van De Ville et al., 2021), pre-processing was performed with a pipeline comprising Global Signal regression (GSR) given its capacity to improve the explanatory value of resting-state FC for behaviour and abolish motion artifacts (Li et al., 2019). GS is hypothesised to contain a complex mixture of non-neuronal artefacts (e.g., physiological, movement, scanner-related) and its removal, while effective, is widely debated in light of differing results for psychedelic effects due to the presence of anticorrelations (Palhano-Fontes et al., 2015; K. H. Preller et al., 2018; K. H. Preller et al., 2018). While the present study sought instead to assess the test-retest stability of functional connectomes and not their directionality, a consensus on the suitability of GSR for pharmaco-imaging (McCulloch et al., 2022) and connectome fingerprinting should be reached.

4.6. Conclusion

In summary, the ritualistic use of ayahuasca is associated with reduced connectome idiosyncrasy, marked by a spatiotemporal reconfiguration of brain connectivity traits. Members of Santo Daime pertain to a culture which emphasises the interaction of a psychoactive sacrament with the interpersonal dynamics of ritualism. Ultimately, it is possible that the synergy of the two that produces the blurred connectome fingerprint presented herein. An important next step is employing task designs to directly assess the moderating role of inter-individual differences for the variability of subjective experiences under psychedelics. Going forwards, by celebrating individual differences in the study of subjective experiences we may be a step closer to producing personalised neural markers of psychedelic effects.

Code availability

All code used for analysis is to be made available on P.Mallaroni's GitHub page (<https://github.com/PabloMallaroni>).

CRediT authorship contribution statement

Pablo Mallaroni: Conceptualization, Investigation, Formal analysis, Methodology, Data curation, Writing – original draft. **Natasha L. Mason:** Conceptualization, Investigation, Supervision, Writing – review & editing. **Lilian Kloft:** Conceptualization, Investigation, Writing – review & editing. **Johannes T. Reckweg:** Conceptualization, Investigation, Writing – review & editing. **Kim van Oorsouw:** Conceptualization, Project administration, Resources. **Stefan W. Toennes:** Formal analysis, Writing – review & editing. **Hanna M. Tolle:** Methodology, Validation, Writing – review & editing. **Enrico Amico:** Conceptualization, Methodology, Software, Supervision, Writing – review & editing. **Johannes G. Ramaekers:** Conceptualization, Investigation, Data curation, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

None of the authors declare any conflict of interest.

Data availability

The connectomes and the accompanying covariates used to differentiate individuals can be made available to qualified research institutions upon reasonable request to J.G.R and a data use agreement executed with Maastricht University.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.neuroimage.2023.120480](https://doi.org/10.1016/j.neuroimage.2023.120480).

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